Ger J Exerc Sport Res
https://doi.org/10.1007/s12662-026-01109-5
Received: 2 November 2024
Accepted: 18 March 2026

Carla Luttmann · Petra Jansen
Faculty of Human Sciences, University of Regensburg, Regensburg, Germany

Visual flow speed affects sense of agency but not effort during cycling in virtual reality

Supplementary Information

The online version of this article (https://doi.org/10.1007/s12662-026-01109-5) contains supplementary material, which is available to authorized users.

Introduction

Virtual reality (VR) is a highly immersive medium that is increasingly used in various behavioral contexts, one of which is physical activity. Here, different types of media have shown a positive impact on mood, enjoyment (Jones, Karageorghis, & Ekkekakis, 2014), and engagement (Cao, Peng, & Dong, 2021). Further, visual or auditorial feedback can affect subjective and objective exercise parameters during exercise.

Both of these effects can be observed in VR studies (Cao et al., 2021; Dębska, Polechoński, Mynarski, & Polechoński, 2019; Kraft, Russell, Bowman, Selsor, & Foster, 2011; Lyons et al., 2014; Mestre, Ewald, & Maiano, 2011; Warburton et al., 2009; Yao & Kim, 2019). Virtual reality in general, and head-mounted displays (HMD) in particular, can produce great feelings of presence in the virtual world (Campos, Nusseck, Wallraven, Mohler, & Bülthoff, 2007; Yao & Kim, 2019), which increases the credibility of the visual feedback. By manipulating characteristics of the virtual scene, user experience can be affected through direct action feedback presented in a natural first-person perspective.

The primary focus of this study was to investigate the role of visual flow speed in objective and subjective effort during continuous cycling exercise. The results can contribute to developing virtual environments in the exercise domain. The findings can also advance the usefulness of VR in motivating people to participate in regular physical activity that they would otherwise perceive as unpleasant. Further, athletes can benefit from the potential amelioration of perceived effort, as it allows them to train at higher intensities.

Related work

Virtual reality and physical activity

Virtual reality is highly effective in producing all kinds of illusions and directing attention toward external stimuli. Through the diversion of attention, the gamification of exercise, and the establishment of interesting contexts, the combination of VR and physical activity/exercise has positively affected self-efficacy, enjoyment, arousal, performance, affective attitude toward exercise, and exercise frequency (Cao et al., 2021; Plante, Aldridge, Bogden, & Hanelin, 2003; Rhodes, Warburton, & Bredin, 2009; Zeng, Pope, & Gao, 2017). One mechanism by which VR influences users is the sense of body ownership for the virtual avatar—the feeling of being able to perceive and control the avatar as one's own body (Gallagher, 2000). This embodiment of the avatar and adopting its (assumed) traits and competencies have affected user experience and performance in motor learning (Banakou & Slater, 2017) and during short-time incremental cycling (Kocur et al., 2021). Such bodily illusions require a (fully) visible avatar, either through mirrors or third-person perspectives. Evoking similar effects through a specific scene design would represent a more natural method, enabling a first-person perspective on the virtual scene. Perspective can facilitate embodiment in VR (Gorisse, Christmann, Amato, & Richir, 2017; Maselli & Slater, 2013) even without additional sensory input (Carey, Crucianelli, Preston, & Fotopoulou, 2019) or despite asynchronous tactile stimulation (Maselli & Slater, 2013). Especially for navigation, a first-person perspective can be advantageous because it enables a more familiar perception of the environment (Medeiros et al., 2018).

Visual feedback in virtual reality

Scene properties are generally crucial for the situational context and atmosphere in the virtual environment. Further, the virtual scene provides sensory action feedback such as movement speed or direction. Such cues are typically related to user behavior, which creates a feeling of responsibility for the sensory outcome. The feeling of responsibility for an action is termed *sense of agency* (Pyasik, Furlanetto, & Pia, 2019). Without it (and without action feedback), using VR would barely differ from watching a movie. The sense of agency is generally related to the plausibility of the action feedback and, in VR, it is further coupled to the immersion into the virtual scene (Barbot & Kaufman, 2020). It has been demonstrated that manipulating sensory action outcomes in VR can induce a sense of agency for actions that the user has not executed, for example, for speech (Banakou & Slater, 2017) and movements (Burin et al., 2018). Incongruent sensory feedback, however, coincides with decreased agency (Evans, Gale, Schurger, & Blanke,

Published online: 15 April 2026

2015). Most importantly, virtual action feedback has to be plausible and follow an inherent taxonomy (Gonçalves, 2023).

Although action feedback consists of cues from multiple sensory qualities, visual signals are particularly relevant for motor control because of their fast processing and importance for postural stability (Janeh et al., 2017; Prokop, Schubert, & Berger, 1997). During locomotion, the central visual feedback cue is the *visual flow*. It provides information about movement speed and direction and includes the *optical edge rate* and the *global rate of optical flow* (Mohler, Thompson, Creem-Regehr, Pick, & Warren, 2007). The angular motion of features on the retina of an observer describes the global rate of optical flow. It varies with the distance and direction of those features relative to the observer. This is especially relevant for studies on speed perception because gaze direction largely affects visual flow characteristics. For instance, the global flow rate is higher in the periphery (*lamellar flow*) than in the central flow (near the focus of expansion; Banton, Stefanucci, Durgin, Fass, & Proffitt, 2005; Campos et al., 2007). In HMD, the area of lamellar flow is reduced due to the smaller visual field. Consequently, movement speed in VR is often underestimated (Jörges & Harris, 2022). Visual flow is further affected by the optical edge rate—the density of visual features passing a reference point. A common example where optical edge rate is used for movement speed control is the distance of guideposts on highways (Garach, Calvo, & de Oña, 2022; Iio, Nakai, & Usui, 2023).

Action feedback is continuously processed and integrated into motor control to evaluate action quality and goal attainment during physical activity (Saunders & Knill, 2003). Manipulated visual flow is thus suspected to affect motor behavior, which has been found in studies on driving, gait, and cycling behavior. For instance, presenting conflicting visual flow (regarding optical edge rate and global rate of optical flow) that does not match the actual movement speed elicits compensational movements during treadmill walking (Ludwig et al., 2018) to keep balance and maintain a constant visual flow rate. Although the actor does not actually produce the visual flow speed, they adapt their stride length or frequency to match the visual flow. Similarly, speed control during car driving is affected by visual flow speed (Pretto & Chatziastros, 2006). During ergometer cycling, Parry, Chinnasamy, and Micklewright (2012) found a decreased perception of effort during time trials with decreased visual flow speed. Participants were asked to match their power output to a previous time trial without VR while seeing matching, increased, and decreased visual flow in immersive VR. Although the task was to match their power output to a previous trial, participants cycled at higher resistances when viewing decreased visual flow compared to matching or increased visual flow, while maintaining the same heart rate and reporting lower perceived effort. This underlines the role of visual flow in performance appraisal and distance estimation. Especially for exercise contexts, where higher self-efficacy has been found to decrease perceived exertion (Hutchinson, Sherman, Martinovic, & Tenenbaum, 2008) and improve affect (Wienke & Jekauc, 2016), such agency illusions seem applicable.

Because most visual flow studies focus on preferred walking speed (Janeh et al., 2017) or driving simulations (Pretto & Chatziastros, 2006), assumptions about the effect of altered visual flow on constant load cycling are ambiguous. In contrast to driving, movement speed during cycling is directly related to physiological effort (pedaling force, pedaling cadence, stride length, etc.). Consequently, visual flow speed illusions may also affect perceived effort during exercise and even alter general perceptions of physical abilities.

In a recent study, it was found that a 20% visual flow speed difference between conditions did not affect heart rate or perceived effort during cycling in immersive VR (Luttmann et al., 2024). This was partly ascribed to a lack of discriminability of the visual flow conditions due to the inter-trial interval of 1 week and the speed differences of 20%. Adapting to the limitations of this study design, the present study looked into larger speed differences and all trials were executed in one session. Also, the matching visual flow speed was assessed individually to ensure that both fast and slow visual flows were perceived as unmatching. We assumed that the visual flow speed provides information about the movement speed and, thus, visualizes the user's athletic abilities. Although heart rate and perceived effort are tightly coupled with the actual physical effort (Borg, 1982), the VR was expected to provide credible performance feedback, altering both measures of effort. We aimed to gain insight into the mechanisms of visual flow speed perception in VR and its impact on physical and perceived effort.

Objectives and hypotheses

Hypothesis 1. Heart rate decreases with increasing visual flow speed.

Hypothesis 2. Due to the tight coupling of heart rate and perceived effort, we hypothesize that perceived effort decreases with increasing visual flow speed.

Exploratory measurements

In addition to these hypotheses, we investigated the effects of visual flow speed on post-exercise self-perception of fitness, sense of agency, sense of ownership, sense of presence, and speed perception. Exploratorily, we analyzed the data for differences between the two experimental conditions. We further conducted exploratory analyses of pedaling cadence and its impact on heart rate and rating of perceived exertion (RPE).

Method

In a repeated measures within-subject design, participants completed two 10-min cycling trials at a constant resistance in a single experimental session. The study was preregistered at OSF (https://osf.io/bprgc/?view_only=3907dfc6c2724f81b29883821f28a51b).

Participants

Based on the effect size $f = 0.19$ in the study by Kocur et al. (2021), a power analysis with G*Power (Faul, Erdfelder, Buchner, & Lang, 2009) yielded a required sample size of $N = 55$ (repeated measures ANOVA, $\alpha = 0.05$, $f = 0.15$, $\beta = 0.80$). In total, 66 university students of applied movement science (34 male, 32 female) were recruited to ensure a balanced distribution across both condition orders and to account for data exclusion. Participants had to confirm they had no acute or chronic cardiovascular or muscular diseases or known sensitivity to motion- and cybersickness.

Virtual reality system

For the VR system, an HTC Vive HMD (HTC Vive, Taoyuan, Taiwan) was used in combination with two HTC Vive trackers attached at the pedals of the ergometer. The VR system ran on a desktop PC with an NVIDIA GeForce® RTX™ 3060 Ti 8 GB GDDR6 graphic card. The virtual scene was programmed in Unity (Version 2020.3.38f1) and presented in VR by Unity and SteamVR (Valve Corporation, Bellevue, WA, USA). The avatars were created in Daz Studio 3D (DAZ Productions, Salt Lake City, UT, USA) and imported into Unity as rigged models. The Final IK Unity Asset (RootMotion, Tartu, Estonia) was used to synchronize the movement of the avatars to the participant's movement.

The virtual scene consisted of a straight road surrounded by trees and mountains (**Fig. 1**). Participants could see parts of a virtual avatar when they looked down on themselves. Similar to a natural cycling situation, participants would see the front tire, the handlebar, and the virtual arms and hands. Looking even further down, they would see the legs and feet, as well as the rest of the virtual bicycle. The avatar was included to enhance the realism of the virtual situation. Based on the participant's gender, the avatar was presented as male or female. This mainly affected hand size, body size, and whether or not the avatar had breasts (**Fig. 2**).

Performance parameters

During cycling, cadence was calculated by the VR system from the rotation time of the right pedal. The displayed cadence was a moving average of the last five values to prevent too-sudden variations. Cadence had to be kept constant at an individual value $\pm$ 5 revolutions per minute (rpm), which was assessed beforehand. Heart rate was measured by a chest strap heartbeat sensor (Polar H10, Polar Electro Oy, Kempele, Finland) measuring continuously and transferring data to a smartphone app (PolarBeat, Polar Electro Oy) via Bluetooth. It was recorded in the last 10 s of every minute. Perceived effort (RPE) was assessed using the Anstrengungsskala Sport (ASS; Büsch, Utesch, & Marschall, 2022) ranging from 1 ("not strenuous") to 10 ("so strenuous that I have to stop") every 3 min. At these time points, the scale was presented in VR, and participants answered verbally without stopping the exercise (**Fig. 3**).

Questionnaire

After each trial, participants completed a computerized questionnaire consisting of the I-group Presence Questionnaire (IPQ; Schubert, Friedmann, & Regenbrecht, 2001), the scales "body ownership" and "agency" of the Body Representation Questionnaire (BRQ; Banakou, Kishore, & Slater, 2018), the scales "strength" and "endurance" of the Physisches Selbstkonzept Skalen (PSK; Stiller, Würth, & Alfermann, 2004) and a question about the suitability of the visual flow speed.

The IPQ consists of the scales "spatial presence," "involvement," "experienced realism," and a general item. Each scale consists of four to five statements that are rated on 7-point scales (−3 to 3). The general item is rated in the same way. The questionnaire aims to assess the subjective experience of feeling present in the virtual environment (Schubert et al., 2001), which affects the effectiveness of VR (Ijaz, Ahmadpour, Wang, & Calvo, 2020). The general item and the spatial presence scale were used to calculate a general presence score.

The BRQ contains five statements on 7-point scales (−3 to 3). The statements "I felt that the virtual body I saw when looking down at myself was my own body" and "I felt that the movements of the virtual body were caused by my own movements" were chosen to assess body ownership and agency (Banakou et al., 2018). Body ownership was expected to remain stable across visual flow speed as the avatar did not change. Differences in agency scores could explain differences in the outcome measures.

Abstract

Ger J Exerc Sport Res
https://doi.org/10.1007/s12662-026-01109-5

C. Luttmann · P. Jansen

Visual flow speed affects sense of agency but not effort during cycling in virtual reality

Abstract
Virtual reality can be a useful tool for exercise facilitation by providing specific action feedback through environment and avatar design. During locomotion, the visual flow speed is a central sensory cue. It conveys information about the actor's movement speed and direction. The impact of visual flow speed manipulations has been predominantly investigated in cycling time trials or comfortable gait, focusing on the effects on pacing and motor control. However, the effects on effort during constant load cycling have rarely been examined. In the present study, 66 university sport students perform two 10-min cycling bouts at a constant, moderate intensity. The speed of the virtual scene corresponded to an individually assessed accurate speed that was either decreased or increased by 50%, respectively. Heart rate and subjective effort were measured throughout each trial. Increased visual flow speed was expected to evoke lower heart rates and ratings of subjective effort. Mixed linear model analyses revealed no significant effects of visual flow speed on either dependent variable. Post-trial questionnaires showed greater senses of agency and presence when cycling with increased visual flow. Increased visual flow speed was further concomitant with slightly lower cadences. Although subjective reports indicated differences in the perception of both exercise bouts, the performance parameters failed to demonstrate an effect of visual flow speed. Adaptation processes to the visual flow speed might account for some of these results.

Keywords
Visual speed perception · False feedback · Action feedback · Locomotion · Exercise facilitation

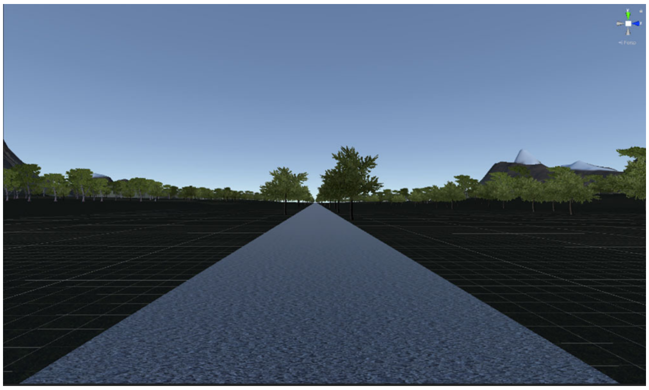

Fig. 1 ▲ Virtual reality scene from participant view

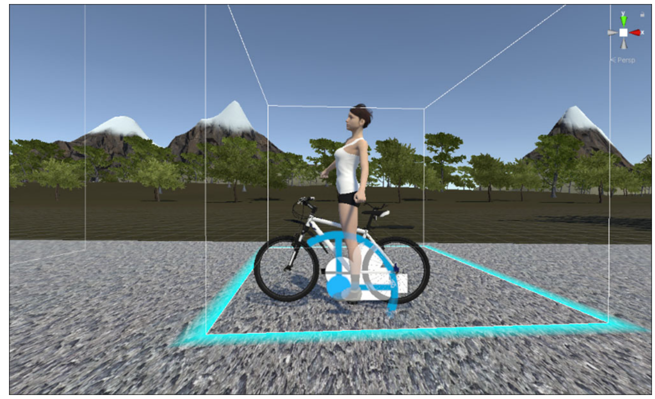

Fig. 2 ▲ Female avatar and bicycle

Fig. 3 ◀ Rating of perceived exertion scale presented every 3 min

The two subscales "strength" and "endurance" from the PSK were included to assess the perception of one's own fitness in the domains of strength and endurance. Each scale contains five statements rated on 4-point scales that are averaged to calculate the scale score (Stiller et al., 2004). These scales were chosen because they are crucial factors for cycling performance. The items assessed whether visual flow speed affects self-perception, which is related to self-efficacy and motivation. Positive effects on self-perceived fitness have been found before for bodily illusions (Kocur et al., 2021).

A question about the match of the visual speed was added to evaluate speed perception and determine possible differences between participants with different perceptions. Visual speed was rated on a 7-point scale from "too slow" to "too fast."

Procedure

Participants received written information about the study and provided their informed consent and demographic information. Then, they were seated on an ergometer (Cyclus2, RBM elektronik-automation GmbH, Leipzig, Germany) at a comfortable saddle height and began cycling at the target resistance to determine their comfortable pedaling cadence. Resistance on the ergometer was kept constant throughout the two trials at 60% of the estimated maximum power output. For male participants, body weight (kg) was multiplied by 3 and for female participants by 2.5 to calculate maximum power output (aus der Fünten, Faude, Skorski, & Meyer, 2013). Next, participants put on the HMD, displaying the virtual scene moving at 5 km/h. Using two keys mounted to the handlebar, participants decreased or increased visual flow speed by 5 km/h increments while cycling at the target cadence and resistance until it subjectively matched their cycling. The visual flow speed of the virtual scene was presented at one of two different levels: individual visual flow speed + 50%, individual visual flow speed – 50%. The order was randomized and counterbalanced between participants. The individual visual flow speed was assessed beforehand to ensure that both experimental visual flow speeds would be perceived as unmatching. This design was chosen to adapt to the limitations of a similar study, where speed differences were assumed to be too small, while the distance between exercise bouts was too large (Luttmann et al., 2024). Due to the expected linear relationship between visual flow speed and the dependent variables, the individual speed was calibrated but not tested to reduce total test time and effort. The setup is shown in ◘ **Fig. 4**.

Participants were familiarized with the ASS on the computer and then seated on the ergometer with the HMD. The experimenter manually adjusted the virtual avatar's positioning so that the body moved realistically on the virtual bicycle. Visual flow speed was adjusted to the appropriate condition for the first trial. When participants were ready to start, the virtual scene was initiated, and the first measures were recorded. Participants completed two 10-min cycling exercises with heart rate being recorded every minute and RPE being recorded every 3 min. After the completion of each trial, participants filled in the computerized questionnaire. The second trial was started after a break of at least 5 min

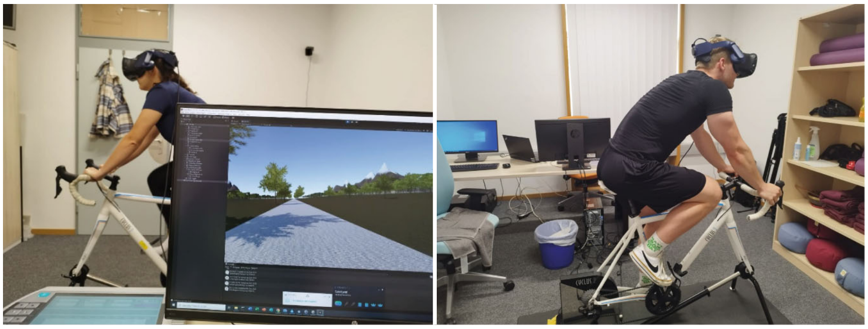

Fig. 4 ▲ Experimental setup

to a maximum of 10 min until the heart rate decreased back to the level before the assessment of the comfortable cadence.

Analysis

Visual flow speed and trials were coded 0 and 1, respectively. All independent variables were then centered on their mean. Separate linear mixed model analyses were performed for each dependent variable using the lme4 package for R (Bates, 2010). All dependent variables were analyzed with the fixed effects "Visual Flow Speed," "Minute," and "Trial." Interindividual differences in the outcome measures between participants were included in all models by adding the random effect "Participant." A maximal model with random slopes for all fixed effects was defined for each measure. Random slopes were reduced stepwise following the procedure of Matuschek, Kliegl, Vasishth, Baayen, and Bates (2017). Complexity reductions were stopped when a likelihood ratio test with $p < 0.2$ indicated a loss in goodness of fit. Subsequently, nonsignificant fixed effects, indicated by a likelihood ratio test with $p < 0.05$, were removed from the model. Assumptions of normality, linearity, homoscedasticity, and autocorrelation were checked visually and formally. Given the robustness of linear mixed models against violations normality, linearity, or homoscedasticity, and the sensitivity to sample size of the respective tests (Knief & Forstmeier, 2021; Schielzeth et al., 2020), the assumptions were only treated as violated when both the plot and the formal test suggested it (Shatz, 2024). In this case, non-parametric tests were performed. The plots can be found in the supplementary material. If autocorrelation was detected in a Breusch–Godfrey test, an ANOVA was used to compare the model without a term for autocorrelation with one including autocorrelation. The model with lower Akaike information criterion and Bayesian information criterion was preferred when the ANOVA was significant ($p < 0.05$). Exploratorily, the questionnaires and the pedaling cadence were analyzed by calculating the respective scale scores and performing individual linear mixed model analyses for each concept. The fixed effects were "Visual Flow Speed" and "Trial." Random slopes for each fixed effect and a random intercept for "Participant" were included. The model reduction procedure was analogous to the outcome measure analyses. Cadence was then included as a fixed effect into the models for heart rate and RPE analysis, following the same complexity reduction procedure.

Results

Six participants were excluded from all analyses because unprecise motion tracking disturbed the correct cadence display. Consequently, data from 60 participants (30 male, 30 female) were analyzed. Each first measure was excluded from statistical analyses as it was uninfluenced by visual flow speed. Quartiles (Q) and interquartile range (IQR) were used to identify outliers, following the boxplot method (Hoaglin, Iglewicz, & Tukey, 1986; Montgomery, 2024). Values below Q1 – 1.5*IQR or above Q3 + 1.5*IQR were removed from the dataset. Complete statistical results are provided in tabular form as supplementary material.

Heart rate

After removing outliers, heart rate averaged at 135.97 ± 17.87 beats per minute (bpm). Values below 84 or above 188 bpm were considered outliers leading to the removal of 18 recordings from two participants from the dataset for heart rate analysis.

Statistical analysis revealed an estimated intercept of 136.75 ± 2.31 bpm. Heart rate increased within the trials, indicated by the significant effect of "Minute" (1.74 ± 0.11 bpm, $\chi^2_{(1)} = 97.21$, $p < 0.001$). Heart rate in the second trial was significantly higher than in the first trial (2.82 ± 0.44 bpm, $\chi^2_{(1)} = 31.94$, $p < 0.001$). Visual flow speed had no effect on heart rate (0.19 ± 0.44 bpm, $\chi^2_{(1)} = 0.19$, $p = 0.662$). **Figure 5** shows the data distribution after outlier removal, separated by visual flow speeds.

Rating of perceived exertion

Results showed that RPE had the same course as heart rate and averaged at 3.90 ± 1.85, suggesting a low-to-medium subjective exertion. No outliers were identified. Statistical analysis of RPE revealed an estimated intercept of 4.76 ± 0.11: RPE increased within the trials, indicated by the significant effect of "Minute" (0.25 ± 0.01, $\chi^2_{(1)} = 248.29$, $p < 0.001$). In the second trial, RPE was significantly higher than in the first trial (0.35 ± 0.06, $\chi^2_{(1)} = 30.84$, $p < 0.001$). Visual flow speed had no effect on RPE (−0.11 ± 0.06, $\chi^2_{(1)} = 3.29$, $p = 0.07$). **Figure 6** shows the data distribution separated by visual flow speeds.

Questionnaire

The answers to the questions about presence, body ownership, agency, and speed perception were transformed to ranges between 0 and 6. **Figure 7** shows all scale means and standard errors separated by condition.

The estimated intercept for the agency was 3.98 ± 0.17. Agency was significantly lower in the second trial (−1.12 ± 0.19, $\chi^2_{(1)} = 27.81$, $p < 0.001$) and significantly higher during increased visual flow (1.09 ± 0.19, $\chi^2_{(1)} = 26.58$, $p < 0.001$). The

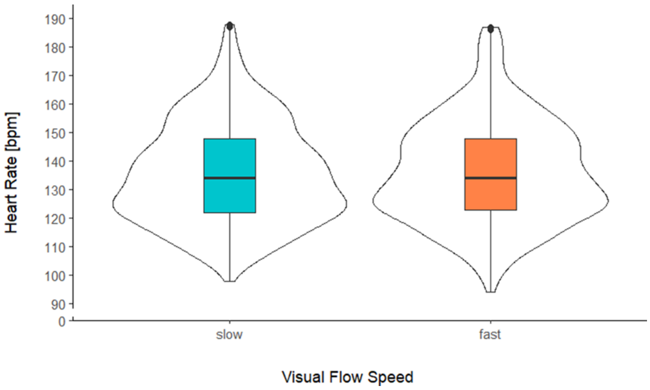

Fig. 5 ▲ Heart rate during the two visual flow speeds. The boxplots display the median as the middle line. The lower and upper outlines of the boxes correspond to the first and third quartiles, respectively. The length of the whiskers corresponds to 1.5*IQR. Dots outside of the whiskers represent outliers. The violin plots display data distribution

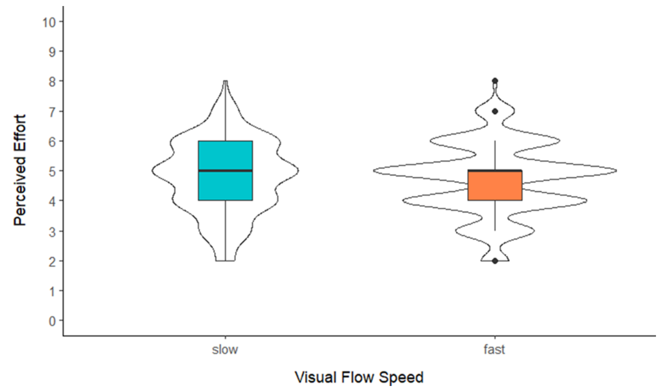

Fig. 6 ▲ Rating of perceived exertion (RPE) separated by visual flow speed. The boxplots display the median as the middle line. The lower and upper outlines of the boxes correspond to the first and third quartiles, respectively. The length of the whiskers corresponds to 1.5*IQR. Dots outside of the whiskers represent outliers. The violin plots display data distribution

estimated intercept for body ownership was 2.13 ± 0.19. Body ownership did not differ significantly between trials (−0.08 ± 0.16, $\chi^2_{(1)} = 0.23$, $p = 0.630$) or visual flow speeds (0.17 ± 0.16, $\chi^2_{(1)} = 1.12$, $p = 0.291$).

The estimated intercept for strength was 2.93 ± 0.07. Strength ratings were significantly higher after the second trial (0.06 ± 0.03, $\chi^2_{(1)} = 4.67$, $p = 0.031$). Visual flow speed did not affect strength ratings significantly (0.05 ± 0.03, $\chi^2_{(1)} = 3.75$, $p = 0.053$). The estimated intercept for endurance was 2.86 ± 0.10. Endurance ratings were significantly higher after the second trial (0.07 ± 0.03, $\chi^2_{(1)} = 6.83$, $p = 0.009$). Visual flow speed did not affect endurance ratings significantly (0.05 ± 0.03, $\chi^2_{(1)} = 3.15$, $p = 0.076$).

Speed perception could not be analyzed with linear mixed models due to a lack of variance between participants. A linear model was calculated with "Visual Flow Speed" and "Trial" as predictors. To account for between-subject variability, participant ID was included as a fixed effect. The effects were tested with Wilcoxon–Rank tests. The virtual speed was perceived as significantly slower during the second trial (0.61 ± 0.16, $p < 0.001$) and significantly faster during increased visual flow (−1.61 ± 0.16, $p < 0.001$).

The estimated intercept for presence was 3.61 ± 0.13. Presence did not differ significantly between trials (−0.19 ± 0.11, $\chi^2_{(1)} = 2.92$, $p = 0.088$). Presence was significantly higher during increased visual flow (0.54 ± 0.11, $\chi^2_{(1)} = 20.03$, $p < 0.001$).

Exploratory analysis of cadence

Effects of visual flow speed on cadence were analyzed exploratorily. Autocorrelation was detected and included in the model. The estimated intercept was 58.84 ± 1.05 rpm. Cadence was significantly affected by visual flow speed (−0.66 ± 0.15, $\chi^2_{(1)} = 32.51$, $p < 0.001$). It was significantly higher during the second trial (0.62 ± 0.15, $\chi^2_{(1)} = 86.04$, $p < 0.001$) and increased within each trial (0.09 ± 0.02, $\chi^2_{(1)} = 80.81$, $p < 0.001$). Because of the significant main effect of visual flow speed on cadence, we included cadence as a main effect in the models for heart rate and RPE analysis. These analyses yielded a significant main effect of cadence on heart rate (0.25 ± 0.08, $\chi^2_{(1)} = 262.01$, $p < 0.001$) and RPE (−0.03 ± 0.01, $\chi^2_{(1)} = 4.18$, $p = 0.047$).

Discussion

The present study aimed to investigate the impact of visual flow speed on the experience of constant load cycling, examining heart rate and RPE during two consecutive exercise bouts in VR. The main hypothesis, that is, increased visual flow evokes lower heart rates and RPE, could not be confirmed. One explanation could be that heart rate during physical activity highly depends on muscular and respiratory effort (Löllgen, 2015). The present results suggest that visual external cues do not sufficiently affect heart rate to withstand the physiological effects of effort. This is in contrast to preceding studies using avatars or visual flow speed manipulations (Kocur et al., 2021) which report significant effects of avatar muscularity on heart rate over time during short incremental cycling exercises. Parry et al. (2012) found no significant effect of visual flow speed on heart rate during cycling. However, their results demonstrate higher-power outputs at decreased visual flow. Consequently, the unaltered heart rate suggests that visual flow speed affects physiological effort. Because resistance was fixed in the present study, we only looked into cadence differences between conditions. The results suggest that faster visual flow leads to lower cadences. However, the difference is of little practical relevance, given that the estimate is close to zero. The significant effect of cadence on heart rate and RPE demonstrates the dependency of physiological measures on muscular effort (Löllgen, 2015). During time trials, as in the study by Parry et al. (2012), movement speed is essential for evaluating goal attainment. In the present study, despite the absence of an action goal, visual flow speed affected cadence without directly affecting heart rate or RPE. Considering the restriction of the cadence range, the effect might be larger in an unrestricted setting. This effect on motor control is similar to studies on

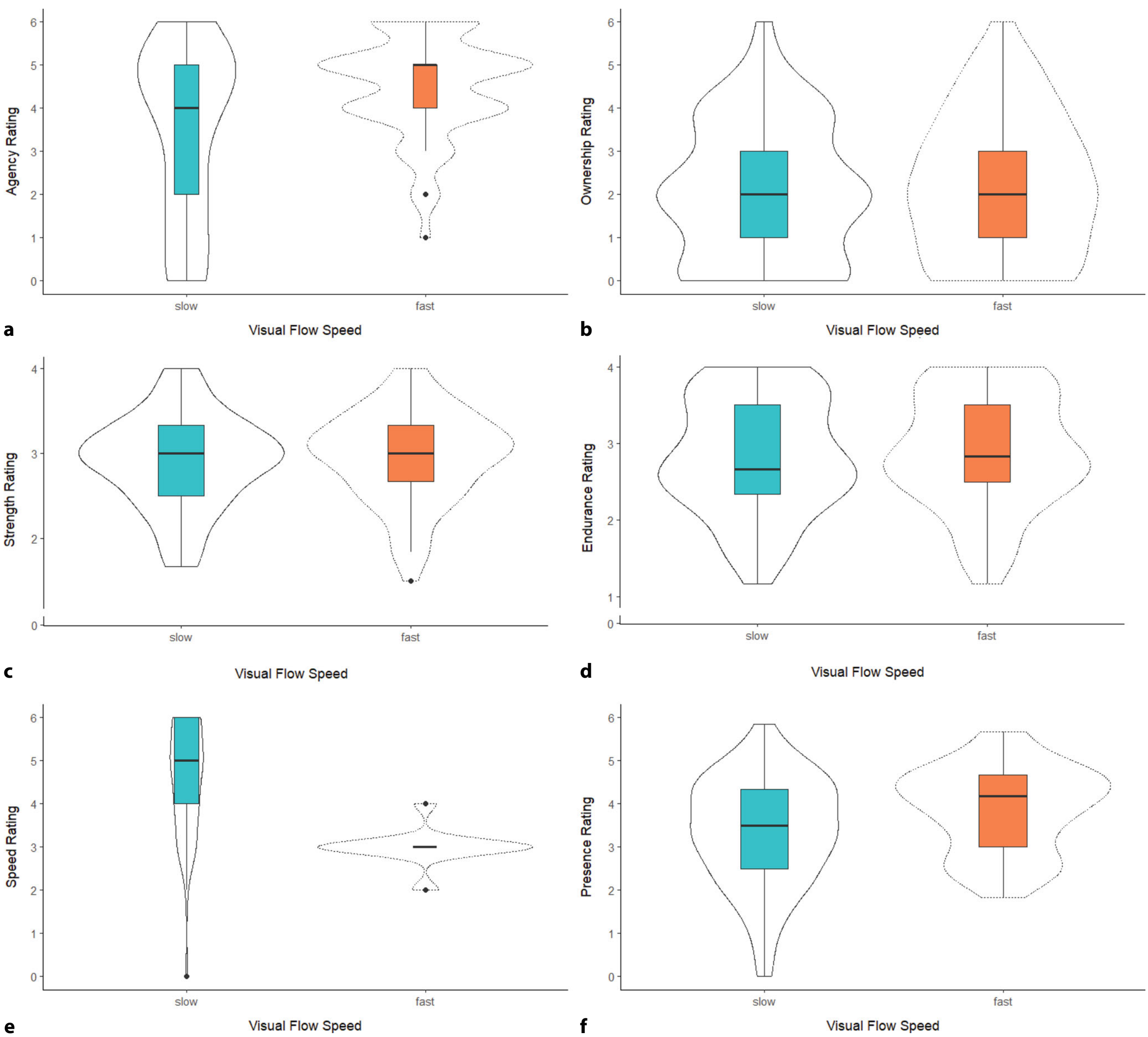

Fig. 7 ▲ Questionnaire scores at the two trials, separated by visual flow speed. The boxplots display the median as the middle line. The lower and upper outlines of the boxes correspond to the first and third quartiles, respectively. The length of the whiskers corresponds to 1.5*IQR. Dots outside of the whiskers represent outliers. The violin plots display data distribution. **a** Agency. **b** Ownership. **c** Self-perceived strength. **d** Self-perceived endurance. **e** Speed rating. **f** Presence

preferred walking speed (Mohler et al., 2007).

Physiological signals of effort strongly influence RPE, which explains the similar courses of heart rate and RPE (Borg, 1982; Rejeski, 1981). However, movement speed was expected to act on RPE comparably to athletic avatars (Kocur et al., 2021). The null effect is especially surprising considering that multiple participants verbally reported that the decreased visual flow condition felt more exhausting or created the feeling of cycling uphill. As with all subjective measures, it is imaginable that participants tried to fulfill expectations about their fitness when rating their perceived effort. They potentially matched their ratings in the two trials, knowing the objective intensity was constant. The significant effect of "Trial" stands against this explanation. As for heart rate, visual flow speed seems to be an insufficient cue to evoke measurable changes in perception. Nonetheless, the results support the assumed coupling of RPE and heart rate (Borg, 1982). Then again, the significant differences between cadences in both visual flow conditions suggest that visual flow speed was used for motor control. We assumed that visual flow speed provides direct performance feedback that would affect performance parameters. Such an effect was not found regarding heart rate and RPE. Only cadence was affected by the visual flow speed, revealing only small differences in physiological effort. Although these affected heart rate and RPE, the estimated differences are minimal.

The questionnaire data show that visual flow speed affected the perceived realism of virtual locomotion. The significant effect of visual flow speed on agency, paired with the rating of increased flow as more matching, suggests that plausibility is a key factor for agency. This aligns with the assumptions of the comparator model (Blakemore, Frith, & Wolpert, 1999), positing that expected and actual action outcomes need to match to evoke a sense of agency. Recent studies have demonstrated that the relevance of internal and external cues for agency depends on the reliability of the feedback. That is, when motor signals and sensory feedback are incongruent, agency is mainly based on goal-related cues instead of the action–feedback association (Wen, Yamashita, & Asama, 2015). These findings support the relation of agency and speed perception found in the present study and further suggest that action–feedback coupling might not be an absolute prerequisite for inducing agency in VR-based exercise.

Altogether, the present results merely hint at a relation between visual flow speed and subjective and objective effort during cycling in VR. The connection between visual flow speed, cadence, and heart rate/RPE should be investigated more thoroughly. For this, physical and virtual speed need to be coupled.

Limitations

The major limitation of the present design is that the visual flow speed was not coupled with cycling cadence. This would improve realism and feelings of agency for the movement speed. Although matching action–feedback associations are relevant for the sense of agency, it has also been suggested that unreliable internal cues lead to favoritism of external signals, specifically performance feedback (Wen et al., 2015), which enables agency formation despite unmatching action feedback. More generally, the scene was not very complex or engaging. This was important to maintain a straight view on the street and keep visual flow similar between participants. Further, scene characteristics such as object density or scale could affect the immersion into the scene or increase the senses of presence and agency. However, they were not relevant for the present investigation of visual flow speed differences. The homogeneity of the sample further limited the study as all participants were students of applied movement science and possibly more accustomed to cycling than other populations. However, their studies mainly include theoretical teachings of movement science and psychology.

Conclusion

The present study analyzed the effects of visual flow speed manipulation on subjective (RPE) and objective (HR) measures of continuous, moderate-intensity cycling. Following previous studies on avatar design, agency illusions, and visual flow in virtual reality, the main goal was to identify the impact of visual flow speed differences on exercise behavior and perception. In summary, the present results present no significant effect of visual flow speed on HR and RPE. Solely the sense of agency was significantly improved during the fast visual flow condition. Considering that multiple preceding studies identified effects on either physiological or perceived effort during walking and cycling, this result was unexpected. Possible reasons include an insufficient illusion due to missing realism of the virtual scene, the independence of pedaling cadence and visual flow speed, and the difficulty of influencing such robust and effort-dependent parameters like HR and RPE. In total, visual flow speed showed no relevant impact on the cycling exercise. Future studies should examine the limitations and strengthen the methodology.

Corresponding address

Carla Luttmann
Faculty of Human Sciences, University of Regensburg
Universitätsstraße 31, 93053 Regensburg, Germany
carla.luttmann@ur.de

Funding. No funding was granted.

Author Contribution. *Carla Luttmann:* Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing—Original Draft, Writing—Review & Editing. *Petra Jansen:* Writing—Review & Editing, Supervision.

Funding. Open Access funding enabled and organized by Projekt DEAL.

Data availability statement. The data from this study can be retrieved at: https://osf.io/7scqr/overview.

Declarations

Conflict of interest. C. Luttmann and P. Jansen declare that they have no competing interests.

The Ethic Research Board of the University of Regensburg reviewed and approved this study involving human participants. The participants provided their written informed consent to participate in this study.

References

Banakou, D., & Slater, M. (2017). Embodiment in a virtual body that speaks produces agency over the speaking but does not necessarily influence subsequent real speaking. *Scientific Reports*, *7*(1), 14227. https://doi.org/10.1038/s41598-017-14620-5.

Banakou, D., Kishore, S., & Slater, M. (2018). Virtually being einstein results in an improvement in cognitive task performance and a decrease in age bias. *Frontiers in Psychology*, *9*, 917. https://doi.org/10.3389/fpsyg.2018.00917.

Banton, T., Stefanucci, J., Durgin, F., Fass, A., & Proffitt, D. (2005). The perception of walking speed in a virtual environment. *Presence: Teleoperators and Virtual Environments*, *14*(4), 394–406. https://doi.org/10.1162/105474605774785262.

Barbot, B., & Kaufman, J.C. (2020). What makes immersive virtual reality the ultimate empathy machine? Discerning the underlying mechanisms of change. *Computers in Human Behavior*, *111*, 106431. https://doi.org/10.1016/j.chb.2020.106431.

Bates, D. (2010). *lme4: Mixed-effects modelling with R*. https://doi.org/10.18637/jss.v067.i01.

Blakemore, S.J., Frith, C.D., & Wolpert, D.M. (1999). Spatio-temporal prediction modulates the perception of self-produced stimuli. *Journal of Cognitive Neuroscience*, *11*(5), 551–559. https://doi.org/10.1162/089892999563607.

Borg, G. A. V. (1982). Psychophysical bases of perceived exertion. *Medicine & Science in Sports & Exercise*, *14*(5), 377–381. https://doi.org/10.1249/00005768-198205000-00012.

Burin, D., Pyasik, M., Ronga, I., Cavallo, M., Salatino, A., & Pia, L. (2018). "as long as that is my hand, that willed action is mine": timing of agency triggered by body ownership. *Consciousness and Cognition*, *58*, 186–192. https://doi.org/10.1016/j.concog.2017.12.005.

Büsch, D., Utesch, T., & Marschall, F. (2022). Entwicklung und Evaluation der Anstrengungsskala Sport. *German Journal of Exercise and Sport Research*, *52*(1), 173–178. https://doi.org/10.1007/s12662-021-00757-z.

Campos, J. L., Nusseck, H. G., Wallraven, C., Mohler, B. J., & Bülthoff, H. H. (2007). Visualization and (mis) perceptions in virtual reality. In *10. Workshop Sichtsysteme: Visualisierung in der Simulationstechnik*. https://pure.mpg.de/pubman/faces/viewitemoverviewpage.jsp?itemid=item_1790151.

Cao, L., Peng, C., & Dong, Y. (2021). Ellic's Exercise Class: promoting physical activities during exergaming with immersive virtual reality. *Virtual Reality*, *25*(3), 597–612. https://doi.org/10.1007/s10055-020-00477-z.

Carey, M., Crucianelli, L., Preston, C., & Fotopoulou, A. (2019). The effect of visual capture towards subjective embodiment within the full body illusion. *Scientific Reports*, *9*(1), 2889. https://doi.org/10.1038/s41598-019-39168-4.

Dębska, M., Polechoński, J., Mynarski, A., & Polechoński, P. (2019). Enjoyment and intensity of physical activity in Immersive virtual reality performed on innovative training devices in compliance with recommendations for health. *International Journal of Environmental Research and Public Health*, *16*(19), 3673. https://doi.org/10.3390/ijerph16193673.

Evans, N., Gale, S., Schurger, A., & Blanke, O. (2015). Visual feedback dominates the sense of agency for brain-machine actions. *PloS One*, *10*(6), e130019. https://doi.org/10.1371/journal.pone.0130019.

Faul, F., Erdfelder, E., Buchner, A., & Lang, A.-G. (2009). Statistical power analyses using G*power 3.1: tests for correlation and regression analyses. *Behavior Research Methods*, *41*(4), 1149–1160. https://doi.org/10.3758/BRM.41.4.1149.

aus der Fünten, K., Faude, O., Skorski, S., & Meyer, T. (2013). Sportmedizin. In A. Güllich & M. Krüger (Eds.), *Sport* (pp. 171–210). Springer. https://doi.org/10.1007/978-3-642-37546-0_7.

Gallagher, S. (2000). Philosophical conceptions of the self: implications for cognitive science. *Trends in Cognitive Sciences*, *4*(1), 14–21. https://doi.org/10.1016/S1364-6613(99)01417-5.

Garach, L., Calvo, F., & de Oña, J. (2022). The effect of widening longitudinal road markings on driving speed perception. *Transportation Research Part F: Traffic Psychology and Behaviour*, *88*, 141–154. https://doi.org/10.1016/j.trf.2022.05.021.

Gonçalves, G., Coelho, H., Monteiro, P., Melo, M., & Bessa, M. (2023). Systematic Review of Comparative Studies of the Impact of Realism in Immersive Virtual Experiences. ACM Computing Surveys, 55(6), 1–36. https://doi.org/10.1145/3533377

Gorisse, G., Christmann, O., Amato, E. A., & Richir, S. (2017). First- and third-person perspectives in Immersive virtual environments: presence and performance analysis of embodied users. *Frontiers in Robotics and AI*, *4*, Article 33. https://doi.org/10.3389/frobt.2017.00033.

Hoaglin, D.C., Iglewicz, B., & Tukey, J.W. (1986). Performance of some resistant rules for outlier labeling. *Journal of the American Statistical Association*, *81*(396), 991–999. https://doi.org/10.1080/01621459.1986.10478363.

Hutchinson, J.C., Sherman, T., Martinovic, N., & Tenenbaum, G. (2008). The effect of manipulated self-efficacy on perceived and sustained effort. *Journal of Applied Sport Psychology*, *20*(4), 457–472. https://doi.org/10.1080/10413200802351151.

Iio, K., Nakai, H., & Usui, S. (2023). Effects of speed reduction marking patterns on simulated driving speed and lane position. *Transportation Research Record: Journal of the Transportation Research Board*, *2677*(2), 880–897. https://doi.org/10.1177/03611981221108979.

Ijaz, K., Ahmadpour, N., Wang, Y., & Calvo, R. A. (2020). Player experience of needs satisfaction (PENS) in an Immersive virtual reality exercise platform describes motivation and enjoyment. *International Journal of Human–Computer Interaction*, *36*(13), 1195–1204. https://doi.org/10.1080/10447318.2020.1726107.

Janeh, O., Langbehn, E., Steinicke, F., Bruder, G., Gulberti, A., & Poetter-Nerger, M. (2017). Walking in virtual reality: effects of manipulated visual self-motion on walking biomechanics. *ACM Transactions on Applied Perception*, *14*(2), 1–15. https://doi.org/10.1145/3022731.

Jones, L., Karageorghis, C. I., & Ekkekakis, P. (2014). Can high-intensity exercise be more pleasant? Attentional dissociation using music and video. *Journal of Sport and Exercise Psychology*, *36*(5), 528–541. https://doi.org/10.1123/jsep.2013-0251.

Jörges, B., & Harris, L.R. (2022). Object speed perception during lateral visual self-motion. *Attention, Perception & Psychophysics*, *84*(1), 25–46. https://doi.org/10.3758/s13414-021-02372-4.

Knief, U., & Forstmeier, W. (2021). Violating the normality assumption may be the lesser of two evils. *Behavior Research Methods*, *53*(6), 2576–2590. https://doi.org/10.3758/s13428-021-01587-5.

Kocur, M., Habler, F., Schwind, V., Woźniak, P.W., Wolff, C., & Henze, N. (2021). Physiological and perceptual responses to athletic avatars while cycling in virtual reality. In Y. Kitamura, A. Quigley, K. Isbister, T. Igarashi, P. Bjørn & S. Drucker (Eds.), *Proceedings of the 2021 CHI conference on human factors in computing systems* (pp. 1–18). ACM. https://doi.org/10.1145/3411764.3445160.

Kraft, J. A., Russell, W. D., Bowman, T. A., Selsor, C.W., & Foster, G.D. (2011). Heart rate and perceived exertion during self-selected intensities for exergaming compared to traditional exercise in college-age participants. *Journal of Strength and Conditioning Research*, *25*(6), 1736–1742. https://doi.org/10.1519/JSC.0b013e3181e06f13.

Löllgen, H. (2015). Herzfrequenz und Blutdruck. In J. Niebauer (Ed.), *Sportkardiologie* (pp. 87–105). Berlin Heidelberg: Springer. https://doi.org/10.1007/978-3-662-43711-7_9.

Ludwig, C. J. H., Alexander, N., Howard, K. L., Jedrzejewska, A. A., Mundkur, I., & Redmill, D. (2018). The influence of visual flow and perceptual load on locomotion speed. *Attention, Perception & Psychophysics*, *80*(1), 69–81. https://doi.org/10.3758/s13414-017-1417-3.

Luttmann, C., Mayer, M., Siebertz, M., Jost, L., Henze, N., & Jansen, P. (2024). Effects of visual flow velocity on cycling experience in virtual reality. *German Journal of Exercise and Sport Research*. https://doi.org/10.1007/s12662-024-00964-4.

Lyons, E. J., Tate, D. F., Ward, D. S., Ribisl, K. M., Bowling, J. M., & Kalyanaraman, S. (2014). Engagement, enjoyment, and energy expenditure during active video game play. *Health Psychology*, *33*(2), 174–181. https://doi.org/10.1037/a0031947.

Maselli, A., & Slater, M. (2013). The building blocks of the full body ownership illusion. *Frontiers in Human Neuroscience*, *7*, 83. https://doi.org/10.3389/fnhum.2013.00083.

Matuschek, H., Kliegl, R., Vasishth, S., Baayen, H., & Bates, D. (2017). Balancing Type I error and power in linear mixed models. *Journal of Memory and Language*, *94*, 305–315. https://doi.org/10.1016/j.jml.2017.01.001.

Medeiros, D., dos Anjos, R.K., Mendes, D., Pereira, J.M., Raposo, A., & Jorge, J. (2018). Keep my head on my shoulders! In S.N. Spencer, S. Morishima, Y. Itoh, T. Shiratori, Y. Yue & R.W. Lindeman (Eds.), *Proceedings of the 24th ACM Symposium on Virtual Reality Software and Technology* (pp. 1–10). ACM. https://doi.org/10.1145/3281505.3281511.

Mestre, D.R., Ewald, M., & Maiano, C. (2011). Virtual reality and exercise: behavioral and psychological effects of visual feedback. *Studies in Health Technology and Informatics*, *167*, 122–127.

Mohler, B. J., Thompson, W. B., Creem-Regehr, S. H., Pick, H. L., & Warren, W. H. (2007). Visual flow influences gait transition speed and preferred walking speed. *Experimental Brain Research*, *181*(2), 221–228. https://doi.org/10.1007/s00221-007-0917-0.

Montgomery, R.M. (2024). *Techniques for outlier detection: a comprehensive view*. https://doi.org/10.20944/preprints202410.1603.v1.

Parry, D., Chinnasamy, C., & Micklewright, D. (2012). Optic flow influences perceived exertion during cycling. *Journal of Sport and Exercise Psychology*, *34*(4), 444–456. https://doi.org/10.1123/jsep.34.4.444.

Plante, T.G., Aldridge, A., Bogden, R., & Hanelin, C. (2003). Might virtual reality promote the mood benefits of exercise? *Computers in Human Behavior*, *19*(4), 495–509. https://doi.org/10.1016/S0747-5632(02)00074-2.

Pretto, P., & Chatziastros, A. (2006). Changes in optic flow and scene contrast affect the driving speed. In *Driving simulation conference europe (DSC europe 2006)* (pp. 263–272).

Prokop, T., Schubert, M., & Berger, W. (1997). Visual influence on human locomotion Modulation to changes in optic flow. *Experimental Brain Research*, *1*(114), 63–70. https://www.infona.pl/resource/bwmeta1.element.springer-360fb604-f04e-34d9-bbc5-d8163d03d64a.

Pyasik, M., Furlanetto, T., & Pia, L. (2019). The role of body-related afferent signals in human sense of agency. *Journal of Experimental Neuroscience*. https://doi.org/10.1177/1179069519849907.

Rejeski, W.J. (1981). The perception of exertion: a social psychophysiological integration. *Journal of Sport Psychology*, *3*(4), 305–320. https://doi.org/10.1123/jsp.3.4.305.

Rhodes, R.E., Warburton, D.E.R., & Bredin, S.S.D. (2009). Predicting the effect of interactive video bikes on exercise adherence: an efficacy trial. *Psychology, Health & Medicine*, *14*(6), 631–640. https://doi.org/10.1080/13548500903281088.

Saunders, J. A., & Knill, D. C. (2003). Humans use continuous visual feedback from the hand to control fast reaching movements. *Experimental Brain Research*, *152*(3), 341–352. https://doi.org/10.1007/s00221-003-1525-2.

Schielzeth, H., Dingemanse, N. J., Nakagawa, S., Westneat, D. F., Allegue, H., Teplitsky, C., Réale, D., Dochtermann, N. A., Garamszegi, L. Z., & Araya-Ajoy, Y. G. (2020). Robustness of linear mixed-effects models to violations of distributional assumptions. *Methods in Ecology and Evolution*, *11*(9), 1141–1152. https://doi.org/10.1111/2041-210X.13434.

Schubert, T. W., Friedmann, F., & Regenbrecht, H. T. (2001). The experience of presence: factor analytic insights. *Presence: Teleoperators and Virtual Environments*, *10*(3), 266–281. https://doi.org/10.1162/105474601300343603.

Shatz, I. (2024). Assumption-checking rather than (just) testing: the importance of visualization and effect size in statistical diagnostics. *Behavior Research Methods*, *56*(2), 826–845. https://doi.org/10.3758/s13428-023-02072-x.

Stiller, J., Würth, S., & Alfermann, D. (2004). Die Messung des physischen Selbstkonzepts (PSK). *Zeitschrift für Differentielle und Diagnostische Psychologie*, *25*(4), 239–257. https://doi.org/10.1024/0170-1789.25.4.239.

Warburton, D. E. R., Sarkany, D., Johnson, M., Rhodes, R. E., Whitford, W., Esch, B. T. A., Scott, J. M., Wong, S. C., & Bredin, S. S. D. (2009). Metabolic requirements of interactive video game cycling. *Medicine and Science in Sports and Exercise*, *41*(4), 920–926. https://doi.org/10.1249/MSS.0b013e31819012bd.

Wen, W., Yamashita, A., & Asama, H. (2015). The sense of agency during continuous action: performance is more important than action-feedback association. *PloS One*, *10*(4), e125226. https://doi.org/10.1371/journal.pone.0125226.

Wienke, B., & Jekauc, D. (2016). A qualitative analysis of emotional facilitators in exercise. *Frontiers in Psychology*, *7*, 1296. https://doi.org/10.3389/fpsyg.2016.01296.

Yao, S., & Kim, G. (2019). The effects of immersion in a virtual reality game: presence and physical activity. In X. Fang (Ed.), *HCI in Games*. Lecture notes in computer science, (Vol. 11595, pp. 234–242). Springer. https://doi.org/10.1007/978-3-030-22602-2_18.

Zeng, N., Pope, Z., & Gao, Z. (2017). Acute effect of virtual reality exercise bike games on college students' physiological and psychological outcomes. *Cyberpsychology, Behavior and Social Networking*, *20*(7), 453–457. https://doi.org/10.1089/cyber.2017.0042.